Prior Emergency Department Utilization and Hospital Admission: Variation by Triage Acuity and Insurance Status

ChatGPT

Abstract

Background. Previous emergency department (ED) utilization may provide information about health care needs beyond the severity assigned at the current encounter. Whether the association between prior utilization and hospital admission differs across triage acuity and insurance categories remains insufficiently characterized in analyses of publicly available clinical encounter data.

Methods. We conducted a secondary, cross-sectional analysis of the de-identified Yale ED admission dataset originally assembled from three EDs during 2014–2017. The outcome was admission rather than discharge at the index encounter, and the exposure was the number of ED visits during the preceding year. Multivariable logistic regression represented prior utilization flexibly using natural cubic splines and included interactions with Emergency Severity Index (ESI) category and insurance group. Subgroup-specific predicted admission probabilities and contrasts comparing three versus zero prior visits were calculated by marginal standardization.

Results. Among 560,486 released encounters, 558,029 (99.56%) were included; 166,022 (29.75%) ended in admission. The interaction of prior utilization with ESI was associated with improved model fit (likelihood-ratio $\chi^2 = 978.70$, 12 degrees of freedom, exploratory $p < 0.001$), as was the interaction with insurance ($\chi^2 = 311.44$, 12 degrees of freedom, exploratory $p < 0.001$). The standardized difference in admission probability comparing three versus zero prior visits was +2.34 percentage points among ESI-3 encounters and +3.27 percentage points among commercially insured encounters. The corresponding difference for Medicare encounters was approximately zero.

Conclusions. Associations between prior ED utilization and current admission varied across clinical and insurance-defined groups. Interpretation is constrained by the absence of patient identifiers in the public release, precluding within-patient correlation adjustments; confidence intervals and hypothesis tests reported here are model-based exploratory quantities. Further validation using patient-linked datasets is required before drawing conclusions about clinical decisions or inequities.

1 Introduction

Emergency departments serve patients with heterogeneous clinical presentations and social circumstances. Admission decisions reflect the assessed severity of the current presentation together with clinical history, anticipated resource needs, local practice, and the availability of follow-up care. The

Emergency Severity Index (ESI) is a five-level triage system in which lower scores indicate greater urgency; prior validation work has shown that its categories are associated with hospitalization and other indicators of acuity (Tanabe et al., 2004; Wuerz et al., 2000). Nevertheless, patients with identical assigned ESI levels can have different patterns of historical healthcare use, comorbidity, and access to care.

Prior ED utilization is an accessible summary of earlier interactions with the healthcare system. Frequent ED use has long attracted interest from healthcare researchers and policy makers, yet the population of frequent users is heterogeneous and cannot appropriately be characterized solely by visit counts (LaCalle & Rabin, 2010). High utilization may reflect chronic illness, repeated acute presentations, unmet outpatient needs, fragmented care, or other determinants. Consequently, prior utilization could have different associations with current hospital admission for patients presenting at different levels of triage urgency.

The dataset analyzed here was originally collected to develop machine-learning models predicting admission at ED triage. Hong et al. (2018) demonstrated that historical utilization information improved the discrimination of admission-prediction models alongside triage variables. In subsequent work, Hong et al. (2019) investigated prediction of short-term ED return among discharged patients. These studies established the predictive value of longitudinal history features in emergency care, but their primary aim was prediction rather than description of how the utilization–admission association varies across clinically and administratively defined subgroups.

Insurance status is another dimension along which hospital use may vary. It can reflect eligibility rules, employment circumstances, age, disability, financial resources, and access to outpatient services, but it is an imperfect proxy for any one of these constructs. Research on ED triage has documented differences in assigned acuity across demographic groups even after observed clinical and contextual adjustment (Patel et al., 2024). Such evidence motivates careful subgroup analyses, not a presumption that any observed insurance-related difference in admission probability represents discrimination. In particular, differences in observed patient composition and unmeasured clinical circumstances remain important alternative explanations.

The primary objective of this study was to characterize the association between prior-year ED visit frequency and current hospital admission, allowing that association to differ by ESI and recorded insurance group. We addressed three questions: (1) how admission probability varied with utilization history, (2) whether the association varied across ESI categories, and (3) whether the association varied across insurance groups. We emphasized adjusted probabilities and percentage-point contrasts rather than only regression coefficients. The analysis is explicitly exploratory because the de-identified release omits stable patient identifiers, preventing correction of encounter-level standard errors for repeated observations from the same patients.

2 Methods

2.1 Study design and data source

This retrospective secondary analysis used the processed, de-identified research dataset released by Hong et al. (2018). The original study sampled adult encounters resulting in admission or discharge at one academic and two community EDs within the Yale New Haven Health system, from March 2014 through July 2017. The public file, `5v_cleandf.RData`, contains 560,486 encounter-level rows and 972 original variables. The current work represents a reanalysis of the same underlying encounters, not a new clinical cohort. No external validation or independent patient sample was available.

The unit of analysis was an *ED encounter*. The public analysis file does not preserve patient and encounter identifiers required to identify multiple rows belonging to the same individual; therefore, the number of unique patients in the analytic subset and within-patient correlation could not be recovered. We followed relevant STROBE reporting principles in describing sample selection, variables, statistical procedures, and sources of bias (von Elm et al., 2007).

2.2 Eligibility, exposure, and outcome

We included released records with an identifiable admission/discharge disposition, a nonnegative prior ED visit count, ESI level 1–5, and an eligible recorded or missing adult age. The analysis code removed 2,457 records (0.44% of the released file), corresponding to missing ESI, leaving 558,029 encounters. Other listed eligibility conditions did not further reduce the sample in the processed analysis. Admission was defined as a disposition label of `Admit`; discharge was `Discharge`. Encounters with other disposition codes were not included in the source analysis release. This binary outcome concerns the current encounter, not subsequent readmission or return to the ED.

The primary exposure was `n_edvisits`, the count of ED visits during the preceding year as constructed in the original data-processing pipeline (Hong et al., 2018). We set $X_i = \log(1 + N_i)$, where N_i is the prior-year visit count, to reduce the influence of extreme visit histories; a three-degree-of-freedom natural cubic spline allowed a nonlinear relationship (Harrell, 2015). Counts were also summarized descriptively in categories of 0, 1, 2, 3–4, 5–9, and 10 or more previous visits. ESI was treated as a categorical variable, with level 3 as the regression reference category; ESI 1 signifies the greatest triage urgency and ESI 5 the least. Insurance was grouped into Medicaid, commercial, Medicare, other, and self-pay categories, following the categories in the released processing file.

2.3 Adjustment variables and missingness

The prespecified modeling implementation adjusted for age (four-degree-of-freedom natural spline), sex/gender recorded in the file, race, ethnicity, arrival mode, ED site, arrival month, and five indicators of recorded clinical history: congestive heart failure, diabetes without complication,

chronic kidney disease, chronic obstructive pulmonary disease, and malignancy. To stabilize estimation, uncommon levels of several categorical variables were pooled into an "Other" category; unrecorded levels were represented as "Unknown" where applicable. Eleven records (0.002% of the released file) lacked age; these were assigned the sample median with a missing-age indicator. Arrival mode was missing in 21,884 released records (3.90%); these were retained using an explicit unknown category. These approaches are pragmatic modeling choices, not a formal missing-at-random analysis.

Clinical history variables represent recorded EHR conditions and should not be equated with confirmed diagnoses at the current visit. The adjustment set captures some measured clinical case mix but is incomplete: in particular, the current fitted specification does not explicitly adjust for current presenting complaint, physiologic instability, all historical diagnoses, or clinician/hospital capacity decisions. No causal interpretation is made from adjusted contrasts.

2.4 Statistical models and tests

We estimated logistic regression models for the probability $\pi_i = P(Y_i = 1 \mid X_i, E_i, I_i, Z_i)$, where Y_i denotes current hospital admission, E_i the ESI category, I_i insurance group, and Z_i the adjustment variables. The most flexible fitted model was

$$\begin{aligned}\operatorname{logit}(\pi_i) = \alpha + f(X_i) + \beta_{E_i} + \gamma_{I_i} + h(A_i) + \boldsymbol{\delta}^{\mathsf{T}} Z_i \\ + \sum_{e \neq 3} \mathbf{1}(E_i = e) f_e(X_i) + \sum_{k \neq \text{Medicaid}} \mathbf{1}(I_i = k) g_k(X_i),\end{aligned} \tag{1}$$

where f, f_e, and g_k represent three-basis natural spline components, h is a four-basis spline of age, and Z_i includes the remaining adjustment terms. Both interactions enter additively on the logistic linear predictor; we did not fit a three-way utilization–ESI–insurance interaction. Thus the fitted model permits nonlinear utilization associations that differ by ESI and by insurance on the conditional log-odds scale. It does not assert that the same conditional associations are causal effects.

We fitted five nested models: a covariate/ESI/insurance baseline, a model adding prior-year utilization, a model with utilization-by-ESI interaction, a model with utilization-by-insurance interaction, and a model containing both interactions. Model comparisons used likelihood-ratio statistics and Akaike's information criterion (AIC). These tests and the model-based confidence intervals assume independent encounter contributions and therefore cannot be considered definitive inferential results given the missing patient identifiers. The large sample also makes very small associations capable of yielding extremely small nominal p values; substantive interpretation emphasizes absolute differences.

2.5 Standardized adjusted probabilities

To summarize association on the probability scale, we estimated model-standardized admission probabilities for $N = 0, 1, 3, 6$ previous ED visits. For each ESI or insurance subgroup g, we retained its observed covariate distribution while setting the utilization value to n in the prediction equation:

$$\widehat{p}_g(n) = \frac{1}{m_g} \sum_{i \in \mathcal{S}_g} \operatorname{expit}\{\widehat{\eta}_i(N = n)\}, \qquad \widehat{\Delta}_g = \widehat{p}_g(3) - \widehat{p}_g(0). \tag{2}$$

The standardization set $\mathcal{S}_g$ comprised up to 20,000 randomly sampled observed encounters per subgroup (all encounters when smaller), with a fixed seed. ESI and insurance groups were standardized to *their own* covariate distributions, so between-group contrasts partly reflect differences in subgroup composition. The contrast of three versus zero prior visits is a model-based comparison at specified exposure values, *not* the effect of intervening to change prior visits. Approximate 95% Wald intervals used the delta method and the logistic model's conventional variance–covariance matrix; they do not incorporate dependence among encounters or uncertainty from the standardization subsampling (Muller & MacLehose, 2014). The reported intervals should thus be treated as exploratory descriptive uncertainty.

2.6 Sensitivity analysis

We fitted an additional model including $\log(1 + \text{number of previous hospital admissions})$, among encounters for which the count of prior admissions was nonnegative and did not exceed the number of previous ED visits. All 558,029 analyzed encounters satisfied this consistency check. Because earlier hospital admissions are a subset of preceding ED visits, this variable may change the interpretation of the exposure association and cannot simply be treated as an independent confounder. We therefore present this analysis as a secondary specification check, not a preferred causal adjustment model. Computation used R logistic regression with the `splines` package; the supplied source script documents variable transformations and reproducibility settings.

3 Results

3.1 Study encounters and crude admission patterns

The released file contained 560,486 ED encounters; the analytic sample contained 558,029, of which 166,022 (29.75%) resulted in hospital admission. The median number of ED visits in the preceding year was one, and the 95th percentile was 14. Within the analytic sample, Medicaid accounted for 194,336 encounters (34.83%), commercial insurance 188,991 (33.87%), Medicare 122,678 (21.98%), other insurance 50,307 (9.01%), and self-pay 1,717 (0.31%). Table 1 summarizes the observed admission proportions.

Crude admission fractions varied sharply by ESI: 85.62% for ESI 1, 55.00% for ESI 2, 29.08%

for ESI 3, 2.22% for ESI 4, and 0.38% for ESI 5. Across recorded insurance groups, they ranged from 6.99% for "Other" to 95.05% for self-pay. The extraordinarily high self-pay admission fraction is not readily clinically interpretable and requires an audit of source coding and selection before insurance comparisons can be used to inform policy. These descriptive admission proportions are visit-weighted, not patient-level probabilities.

Table 1: Observed ED encounters and hospital admissions by triage acuity and insurance group.

Category	Encounters	Admissions	Admission, %
ESI category			
ESI 1	5,271	4,513	85.62
ESI 2	163,534	89,947	55.00
ESI 3	236,229	68,684	29.08
ESI 4	125,003	2,771	2.22
ESI 5	27,992	107	0.38
Insurance group			
Medicaid	194,336	40,516	20.85
Commercial	188,991	55,915	29.59
Medicare	122,678	64,441	52.53
Other	50,307	3,518	6.99
Self-pay	1,717	1,632	95.05
Total	558,029	166,022	29.75

Note. ESI and insurance rows each independently partition the same analytic sample and should not be added together. The self-pay distribution requires further data-quality investigation.

3.2 Model comparison and interaction patterns

Including prior ED utilization in the baseline model reduced AIC from 478,940.04 to 476,502.29. AIC was 475,489.40 with the ESI interaction alone, 476,156.66 with the insurance interaction alone, and 475,201.96 when both interactions were included. In the joint model, adding the utilization–ESI terms conditional on the insurance interaction yielded a likelihood-ratio statistic of $\chi^2 = 978.70$ (12 degrees of freedom); the corresponding insurance interaction conditional on ESI yielded $\chi^2 = 311.44$ (12 degrees of freedom). Nominal p values were 7.10×10^{-202} and 1.85×10^{-59}, respectively. These results describe improvements in fitted model likelihood under an encounter-independence assumption; they are not valid stand-alone evidence of patient-level statistical significance.

3.3 Adjusted associations across triage severity

Table 2 presents the standardized contrast for three versus zero prior ED visits. For ESI 3, the adjusted admission probability increased from 28.44% at zero prior visits to 30.78% at three prior visits, a difference of +2.34 percentage points (naive model-based 95% interval, +1.93 to +2.75). For ESI 1, the analogous estimate was −2.59 percentage points (−4.64 to −0.54); however, admission

probability was already high in this subgroup, and its trajectory was not monotonic across 0, 1, 3, and 6 prior visits. ESI 2 showed a smaller positive contrast (+0.81 percentage points). Absolute differences for ESI 4 and 5 were small because admission was rare. Figure 1 displays the full set of model-standardized probabilities.

3.4 Adjusted associations across insurance groups

The standardized contrast was largest among commercially insured encounters, for which predicted admission probability increased from 28.57% at zero previous visits to 31.84% at three previous visits: +3.27 percentage points (naive model-based 95% interval, +2.85 to +3.70). Corresponding contrasts were +0.47 percentage points for Medicaid, −0.03 for Medicare, +0.67 for the "Other" category, and +0.46 for self-pay. The Medicare interval included zero. The self-pay subgroup remained near a predicted 95% admission probability across utilization values, confirming that this category behaves unusually in the released data. Figure 2 illustrates these patterns. Insurance-specific differences cannot be interpreted as differences in quality of care without a stronger design and more complete measurement of clinical needs.

Table 2: Adjusted admission probability differences: Three versus zero prior-year ED visits.

Subgroup	Risk difference, pp	Model-based 95% interval, pp
ESI category		
ESI 1	−2.59	[−4.64, −0.54]
ESI 2	+0.81	[+0.27, +1.34]
ESI 3	+2.34	[+1.93, +2.75]
ESI 4	+0.52	[+0.33, +0.70]
ESI 5	+0.13	[−0.03, +0.28]
Insurance group		
Medicaid	+0.47	[+0.05, +0.88]
Commercial	+3.27	[+2.85, +3.70]
Medicare	−0.03	[−0.61, +0.55]
Other	+0.67	[+0.15, +1.19]
Self-pay	+0.46	[−1.51, +2.43]

Note. pp = percentage points. Contrasts are computed within each subgroup using its own observed covariate distribution (at most 20,000 sampled encounters per subgroup). Intervals are delta-method intervals from an ordinary encounter-level logistic model; repeated observations from the same patient cannot be identified, so they may substantially understate uncertainty. Entries are associations, not causal effects.

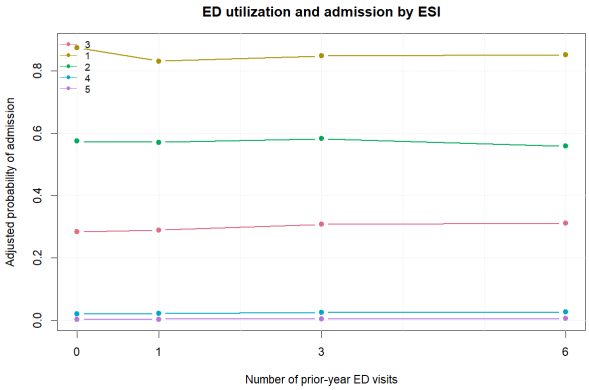


Figure 1: Model-standardized admission probabilities at 0, 1, 3, and 6 previous ED visits, by ESI category.

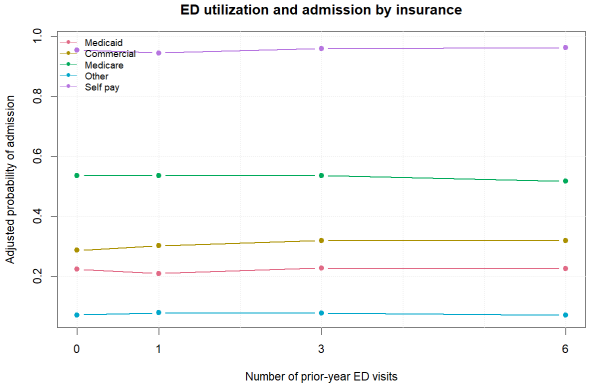


Figure 2: Model-standardized admission probabilities at 0, 1, 3, and 6 previous ED visits, by insurance group.

3.5 Sensitivity analysis for prior hospital admissions

All analyzed records passed the consistency check that the number of preceding hospital admissions did not exceed the number of preceding ED visits. Adding $\log(1 + \text{prior admissions})$ reduced AIC to 460,920.6 on the same analytic sample, relative to 475,202.0 for the model without that term. The added coefficient was 1.104 (conventional standard error 0.0097), indicating that historical inpatient care contains substantial additional information about current admission in this specification. Several utilization-by-ESI and utilization-by-insurance spline coefficients changed after inclusion. Because this variable overlaps structurally with previous ED utilization, and no separately standardized contrasts were computed for the sensitivity model, we do not treat the primary risk differences as proven robust to prior-admission adjustment. A comparison of epidemiological interpretable contrasts under both specifications remains necessary.

4 Discussion

This secondary analysis of more than half a million ED encounters found substantial differences in admission probability across triage acuity and recorded insurance status. In addition, the shape and magnitude of the association between prior-year ED visits and current admission varied across groups after adjustment for selected demographic, clinical-history, temporal, and ED-site variables. The largest positive standardized contrast between three and zero past visits occurred among ESI-3 encounters and commercially insured encounters. These associations were modest on the absolute probability scale when compared with the much larger differences in baseline admission probability across ESI levels.

The findings add a descriptive perspective to earlier prediction-oriented work on this cohort. The original study showed that patient history improves admission prediction (Hong et al., 2018); the present analysis asks whether that history is associated with the outcome in a similar way across clinically defined groups. An interaction in a logistic prediction model is not automatically clinically meaningful heterogeneity. By presenting adjusted probability contrasts, we aim to identify the scale on which differences might matter for patient care or operational planning. Nevertheless, conditional regression interactions and subgroup-standardized probability contrasts are different statistical objects, and subgroup case mix can affect their relationship (Muller & MacLehose, 2014).

ESI 3 comprises an especially informative population for studying heterogeneity because its baseline admission risk is intermediate, permitting meaningful absolute differences in either direction. In contrast, admission was nearly certain for many ESI-1 encounters and very uncommon for ESI-5 encounters. These ceiling and floor effects constrain possible absolute risk differences. A small or negative standardized association in high-acuity encounters should not be interpreted as evidence that frequent users receive less appropriate care. Furthermore, the shape of the association is nonlinear and need not be monotonic, as illustrated by the modeled ESI-2 trajectory.

Insurance-stratified results require even greater caution. The strongest positive contrast was observed for commercial insurance; Medicaid and Medicare estimates were smaller. Differences could reflect underlying age and morbidity distributions, unmeasured presenting complaints, previous access to inpatient services, differences in referral patterns, or data-processing decisions, rather than insurance-dependent admission decisions. Most notably, 95.05% of the encounters labeled self-pay were admissions, an implausibly high figure as a description of general uninsured ED utilization. Before interpreting this group, its raw coding, eligibility, and relation to the released admission/discharge sample should be audited. Recent triage-disparity work establishes the importance of investigating demographic patterns in emergency care but also illustrates the need for nuanced clinical and contextual adjustment (Patel et al., 2024). We therefore avoid equating the present subgroup associations with inequity.

From a health services epidemiology perspective, prior visits may index ongoing health needs rather than an exposure suitable for direct intervention. An observed association between visit history and admission may be useful for describing which encounter types concentrate inpatient

demand, but these analyses cannot determine why an individual sought emergency care, whether the visit was avoidable, or whether admission was necessary. Prior literature has cautioned against simplistic characterizations of frequent ED users (LaCalle & Rabin, 2010). The current contribution should therefore be framed narrowly as exploratory characterization of associations, not development of an admission policy or clinical decision rule.

4.1 Strengths and limitations

A strength is the large, multi-ED encounter dataset with standardized triage scores, utilization history, demographics, and several clinical-history indicators. The fitted model allows nonlinear associations and examines two plausible dimensions of variation, while probability standardization provides an interpretable complement to log-odds coefficients. Full transformation rules and summaries are available in the companion analysis script.

Several limitations materially affect inference. First and most importantly, stable patient identifiers were removed from the public release. Some patients contributed multiple encounters, and conventional likelihood-ratio tests, model-based standard errors, and delta-method intervals assume observations contribute independently. These assumptions are not justified here; nominal significance can be exaggerated, and precise effect-size uncertainty cannot be recovered without linking repeated visits. Neither adjusting for prior utilization nor expanding the number of encounter-level covariates fixes this problem. Second, the dataset was developed for admission prediction and captures only visits ending in admission or discharge at the contributing EDs; it does not represent all possible emergency care dispositions or the broader population. Third, confounding and measurement issues remain: ESI may be imperfect, insurance can be misclassified, and important current clinical factors are omitted from the final model.

These limitations prevent strong claims about differences in patient-level admission propensity or insurance-related fairness. The apparent precision of estimates should not be equated with evidentiary strength. Future studies should use patient-linked encounters to permit cluster-robust or hierarchical inference, check temporal changes and hospital-level heterogeneity, audit insurance/disposition coding, measure presenting complaint and physiologic variables, and evaluate whether the subgroup patterns replicate elsewhere.

5 Conclusion

In a reanalysis of the publicly released Yale ED encounter dataset, the adjusted association between prior-year ED utilization and hospital admission differed across ESI acuity and insurance categories. The largest positive standardized differences between three and zero prior visits were seen among ESI-3 and commercially insured encounters, but absolute differences were generally modest. Because the public data lack patient identifiers, and because one insurance subgroup shows a strikingly unusual admission distribution, the results should be viewed as exploratory rather than definitive

evidence of disparities or treatment differences. Confirmatory epidemiological inference requires a patient-linked, clinically richer dataset.

Declarations

Data availability. The analysis used the publicly released, de-identified dataset associated with Hong et al. (2018): https://github.com/yaleemmlc/admissionprediction. This manuscript does not imply ownership or original collection of those records.

References

Harrell, F. E. (2015). *Regression modeling strategies: With applications to linear models, logistic and ordinal regression, and survival analysis* (2nd ed.). Springer.

Hong, W. S., Haimovich, A. D., & Taylor, R. A. (2018). Predicting hospital admission at emergency department triage using machine learning. *PLOS ONE*, *13*(7), Article e0201016.

Hong, W. S., Haimovich, A. D., & Taylor, R. A. (2019). Predicting 72-hour and 9-day return to the emergency department using machine learning. *JAMIA Open*, *2*(3), 346–352.

LaCalle, E., & Rabin, E. (2010). Frequent users of emergency departments: The myths, the data, and the policy implications. *Annals of Emergency Medicine*, *56*(1), 42–48.

Muller, C. J., & MacLehose, R. F. (2014). Estimating predicted probabilities from logistic regression: Different methods correspond to different target populations. *International Journal of Epidemiology*, *43*(3), 962–970.

Patel, M. D., Lin, P., Cheng, Q., Argon, N. T., Evans, C. S., Linthicum, B., Liu, Y., Mehrotra, A., Murphy, L., & Ziya, S. (2024). Patient sex, racial and ethnic disparities in emergency department triage: A multi-site retrospective study. *The American Journal of Emergency Medicine*, *76*, 29–35.

Tanabe, P., Gimbel, R., Yarnold, P. R., Kyriacou, D. N., & Adams, J. G. (2004). Reliability and validity of scores on the Emergency Severity Index version 3. *Academic Emergency Medicine*, *11*(1), 59–65.

von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Götzsche, P. C., & Vandenbroucke, J. P. (2007). The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: Guidelines for reporting observational studies. *PLOS Medicine*, *4*(10), Article e296.

Wuerz, R. C., Milne, L. W., Eitel, D. R., Travers, D., & Gilboy, N. (2000). Reliability and validity of a new five-level triage instrument. *Academic Emergency Medicine*, *7*(3), 236–242.